* Embedded system → simple Number of Tasks in same Category.

Feedback

10

user

Temperture | -4 | -5 → Computer → Compressor

Desision

Compressor / فریزر

Compressor → Actuator

X *

MultiTasks

Mainprocess

* Computer (PC) Seperate MultiTask

1) Teams 2) Browser 3) Tablet (PC)

Eclipse Embedded systems

PC → Processor, RAm, Monitor, Power supply, ~~GPU~~, ~~Mother Board~~

#

Embedded → Processor, RAM, Power supply, ROM, HDD, SSD

* Micro processor

processor pc

* Micro Controller

ATmega 32A

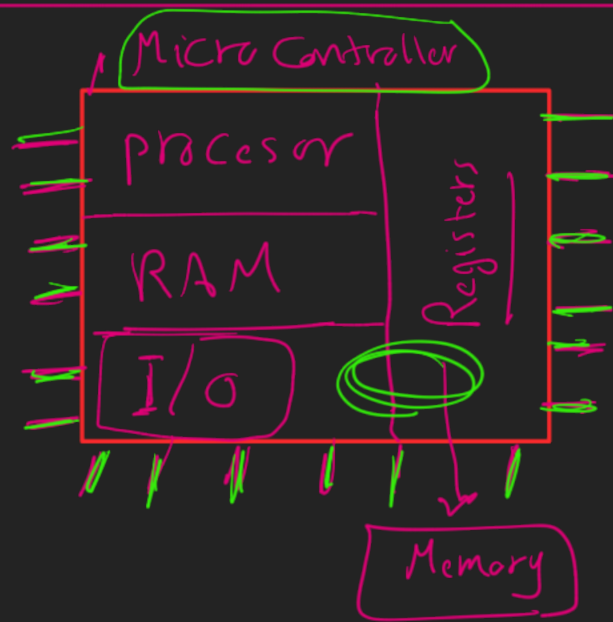
processor

*Can't work alone

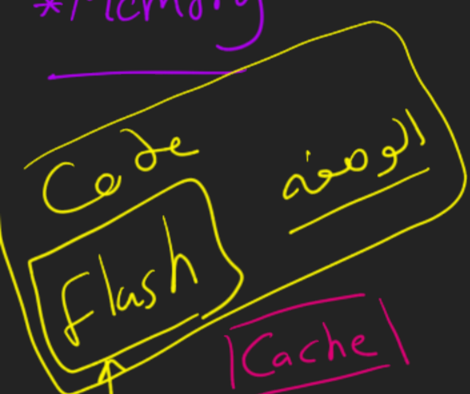
Needs RAM

To work

*Power supply (⚡)



*Memory



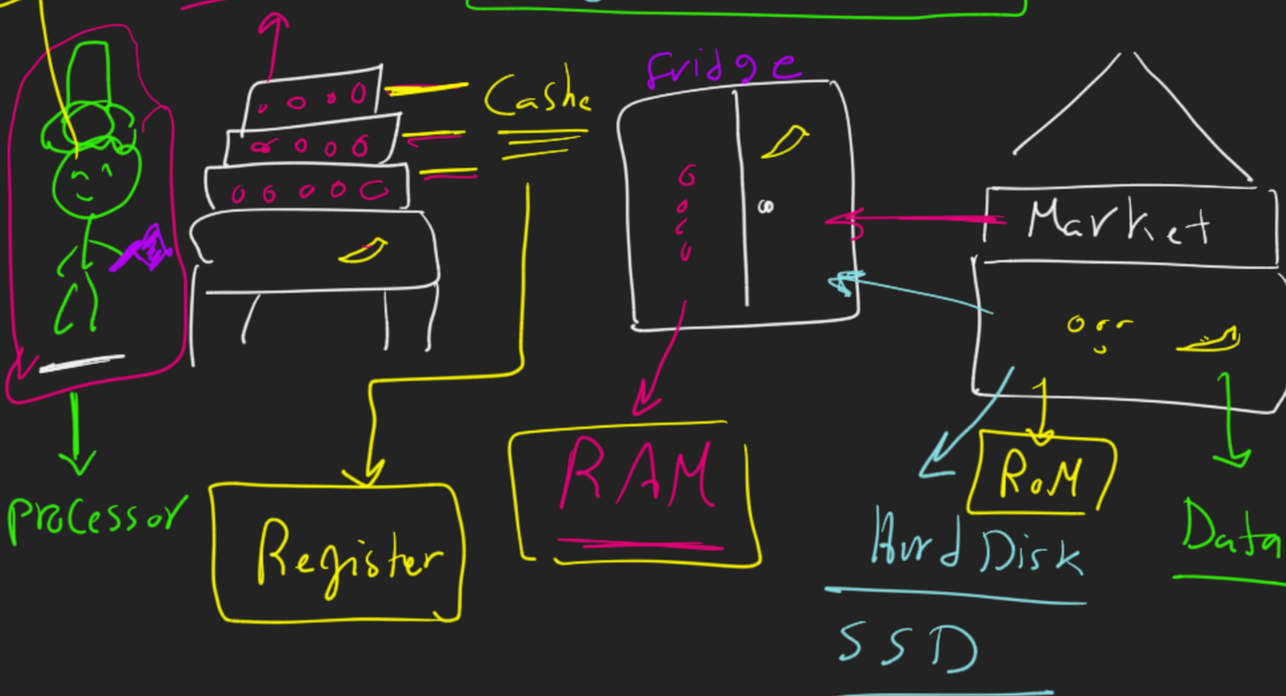
* RAM → SRAM / DRAM

* RoM

* EEPROM / EPROM

* Flash

* Register / cache ✓



	* Volatile Data	* Non volatile Data
Ele on	✓	✓
Ele off	X	✓

* volatile RAM / Cache / Registers

X DRAM → Cap ✓ SRAM → Tr

* Non volatile RoM / EEPRoM / Fash

→ RoM
→ OTP RoM
→ Masked RoM
→ EPRoM
→ EEPRoM
→ Fash RoM

NVRAM

* Hardwar
* Software

	SRAM	DRAM
Cost	✓	
Size		✓
Performance	✓	
Power		✓

CD → OTP

one time programable RoM

EPRoM Ultra Vilot

EEPRoM Electricty

100

EPRoM
EPRoM

Flash RoM

Burn

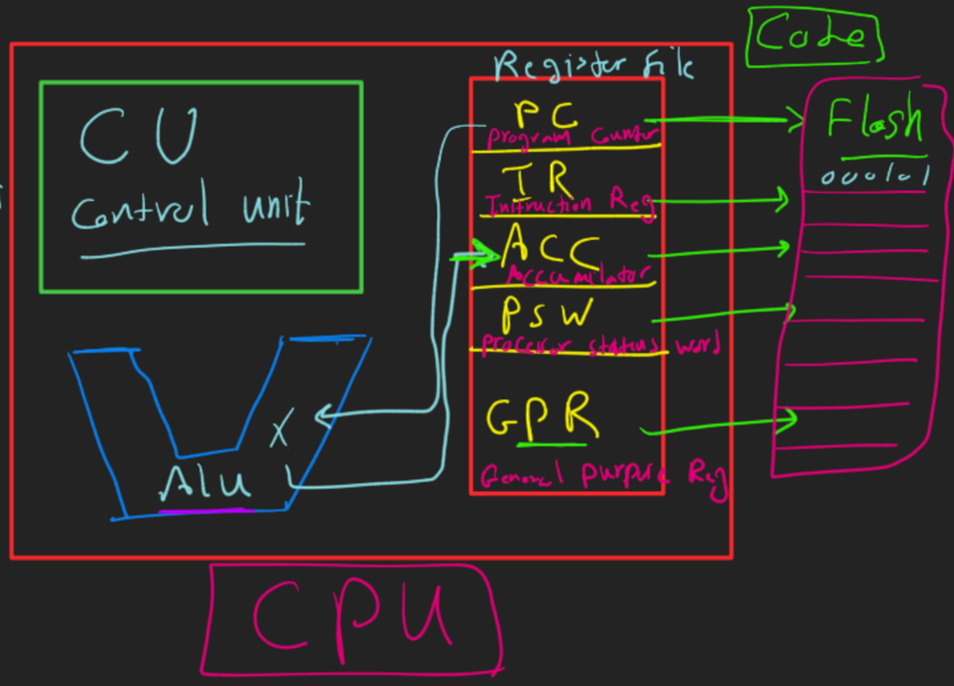
1000,000 / 1000

* processor

CU → Control unit

Reg file

Alu → Arithmatic Logic unit



* Main steps for processor

1) Fetch — Memory
2) Decode — Alu
3) Execute — Control unit

1) fetch

PC → Address Memory

IR → Value Memory

1010100 1

2) Decode : Alu

Instruction format

3 Bits	3 Bits	2 Bits
op code	op 1	op 2
101	010	01
+	2	1

AND 001
OR 010
Sub 110
ADD 101
Shift 111

$= \boxed{3} = \boxed{110}$

3) Execute CU

$\boxed{Acc = 3 = 110}$

0 - 1 / 8 Bits / 2^9

13 $\boxed{-1}$

* Memory (RAM — - RoM —)

* Processor (Alu - CU - Reg)

* fetch - Decode - Excute

C language $\xrightarrow{\text{Compiler}}$ Assembly Language $\xrightarrow{\text{linker}}$ Machine language

#define [ ] [ ] inline

1) Main.c → pre processor → Compiler gcc → Assmbler → Linker → elf → hex

Atmega

* Instruction Set Archtichture

RISC

Reduced
Instraction
Set
Archtichture

CISC

Complex
//
//
//

U32 $X = 5 * 3;$

Processor line

Clock 16MHz

speed of processor

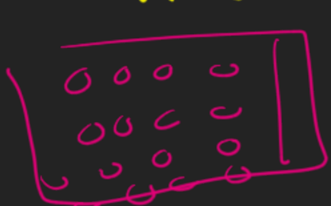
CISC → Complex

4 cycles

single line

RISC → Reduced

4 Lines



each line
1 clock cycle

CISC vs RISC

	CISC	RISC
line	1 line	4 lines
Clock	$\boxed{4}$	$\boxed{1}$ cycle
	$\boxed{4}$	= $\boxed{4}$

8MHz

1000 000 4 cycle

$\boxed{0,5}$ sec

5 hours

Complex

Reduced

* Von Numann Architucture

* Harverd Architucture

$\boxed{\text{Es Archtutur}}$

$$\frac{\cancel{27}}{\cancel{6}} = \boxed{\frac{9}{2}} + \boxed{10 \text{ min}}$$

---

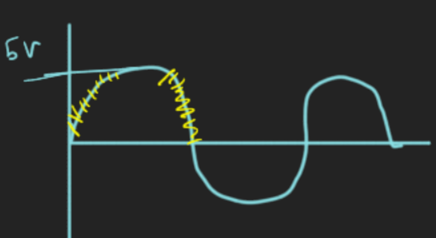
* **DIO** Digital Input output

Atmega32A → Voltge Current

→ 0 → Low | GND | 0V

→ 1 → High | Electricty | 5V

* Analog



* Digital

* DIO Regesters

PIN

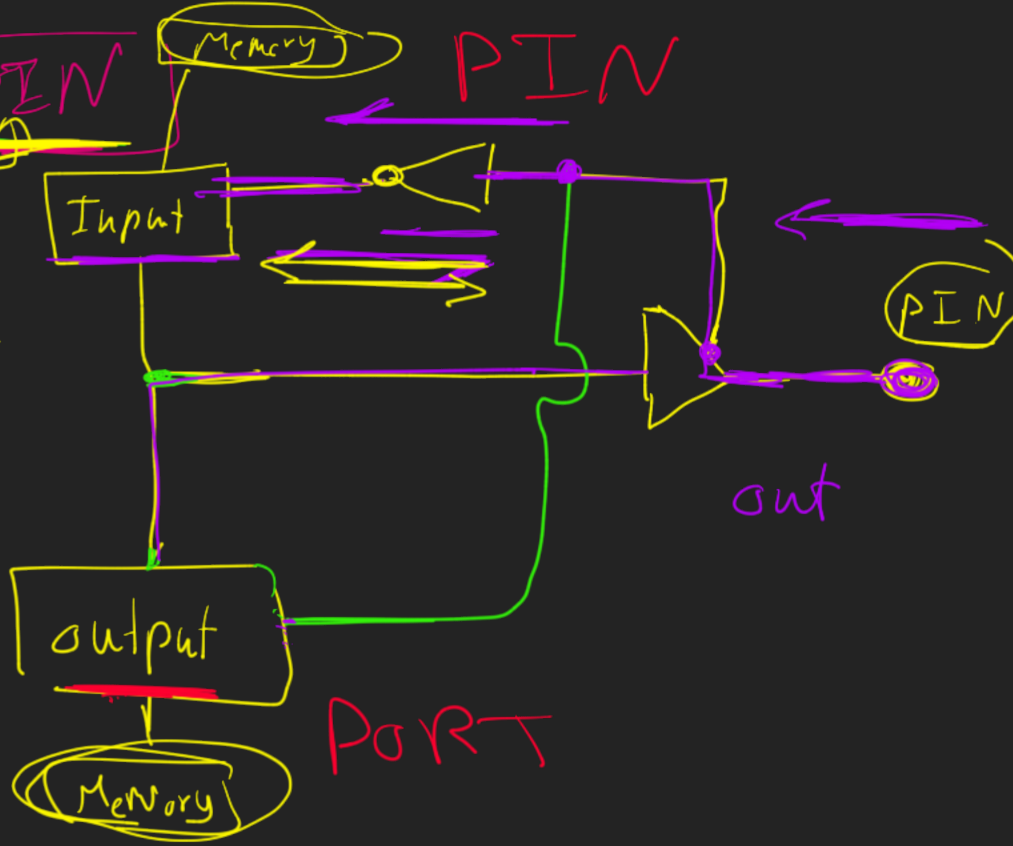
PORT

DIR → 0 → Input → PIN

DIR → 1 → output → PoRT

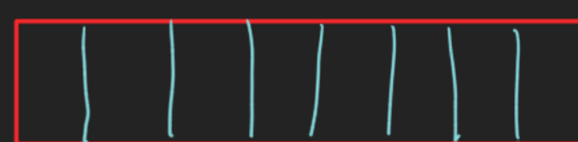
PORT PIN

2) DIR



4 PorTs

8 pins



8 pin

1 Byte

8 Bits

PORTA	DIRA	PINA
PORTB	DIRB	PINB
PORTC	DIRC	PINC
PORTD	DIRD	PIND

```
DIRC = 0000 0100;                out / In
while (1) {
    PORTC = 0000 0100;           8MHz
    delay_ms(1000);
    PORTC = 0000 0000;
    delay_ms(1000);
}
```